MINIFLY RESEARCH / DESIGN PROPOSAL / 01

Continual structural learning

Primary question. Can a locally learned representation keep supporting still-valid relational answers while it adapts to changed transitions? Run the structural test first. Develop the factual-memory comparison independently and integrate the two after both component decisions.

Reference and scope

At archive commit 06c60775, the aligned one-step and two-step model scored 100% on both clean parity holdouts. The primary controls scored 30.03% and 58.51%. Keep that implementation as a positive control. Keep its seven-byte query parser, local readout, generated newline, and one-step and two-step feature families. The parser supplies operand roles. No backpropagation, attention, content-addressed reads, or phase signals to the learner.

Mechanism and three primary arms

Frozen. Deploy the initialized maps throughout. The readout continues learning from the same parity feedback as the other arms.

Direct online. Deploy the current TD maps. For each observed transition of stride one or two, update the source row in the corresponding map with the reference TD rule.

Slow deployment. Perform the same fast TD update, then update only that row of the deployed map by adding tau times the difference between the updated fast row and deployed row. The deployed map supplies the same sparse feature transform and readout. Start both copies from the common initialized maps.

Compute the query features from the currently deployed map before emitting the answer. When teaching feedback arrives, use those pre-answer features for the local readout update. No pair-feature cache may survive a map update unless it is invalidated. Match transition observations, teaching schedule, initialization and readout dimensions across arms. Charge the slow arm for its extra maps and row updates.

Engineering checks and mechanistic attribution

Proposed initial tau is 0.05 per source-row update. Freeze it and all budgets before opening final worlds. On engineering worlds verify that tau = 1 matches the direct arm and that tau = 0 matches frozen deployment. Run a prespecified direct low-step-size control on the same final worlds to distinguish a separate deployed map from simply slowing TD. Use evaluator-only cross-checks with saved maps and readouts to separate representation drift from readout drift. Score those cross-checks separately from live behavior.

One uninterrupted lifetime

Clean, noisy, then changed

Clean. Initialize maps on clean background runs and train the readout on permitted parity pairs. Assign byte identities to positions randomly per world and randomize the answer-byte mapping independently. Audit the clean positive control before continuing.

Noise. Keep the latent cycle and answer rule fixed. As a proposed starting condition, replace 10% of observed background bytes with randomly chosen different in-vocabulary bytes. This disturbs stride-one and stride-two observations without a phase marker. Keep teaching queries clean. Compare with a matched clean continuation on engineering worlds.

Structural change. Swap two byte identities at opposite-parity positions and continue sampling the changed cycle. Keep the answer-byte mapping fixed. Pairs involving exactly one swapped identity change label. Other pairs stay valid. Use the same record format and background-to-teaching schedule across the boundary. No resets, replay, protected-answer flags, or explicit change notification.

Exposure and comparator

Before training, seal a fixed set of ordered pairs and exclude their reverses from feedback throughout the lifetime. Construct balanced scored panels of still-valid and changed pairs, and ensure both categories have examples among the permitted teaching pairs. Retain the original distance holdout as a separate clean reference. Score the same lifetime holdouts before and after change. Report answers from taught pairs separately. An exact or unordered pair table should have no entry for the scored pairs.

Build a transition-count comparator from observed background bytes only. Use a fixed trailing observation budget, infer alternating groups from its counted two-step edges, and learn the answer-byte mapping from allowed feedback. Keep the generator-based oracle as a separate task check. Neither reference feeds the learner.

Score and decide

Use 12 sealed paired worlds. Score first answer plus generated newline at the end of clean and noise phases and at fixed checkpoints after the swap. A proposed post-change schedule is 0, 32, 128 and 512 feedback events, with an equal background budget per event across arms. Report still-valid retention, changed-pair adaptation, both learning curves, exact exposures, runtime and memory. Include each panel size and the per-world contrasts.

Proposed pass: clean joint accuracy at least 85%, noise-phase still-valid accuracy at least 85%, and final changed-pair accuracy at least 85%. Require at least +5 percentage points over direct online on average still-valid accuracy across post-change checkpoints and at least +5 over frozen on final changed pairs. Both paired 95% intervals must exclude zero. Require final changed-pair accuracy within 5 points of direct online. If slow deployment ties the low-step-size control, report support for slower updates without attributing the benefit to the second map.

Combine the factual-memory gains

What has been established

At load 512, the tested fixed-lag float16 store matched float64 at the measured endpoints, with 93.16% old recall and 93.75% unseen-alias recall after clutter. These scores refer to the 64 untouched oldest-wave keys. The 100% revision score was measured immediately after exposure eight. The readout used 32 MiB and fixed expansion arrays added 12 MiB before cache and overhead. Float16 is persistent storage with float64 update arithmetic. The 128-fact setting did not reach 90% old recall after clutter.

Budget-matched comparison

Test a two-by-two comparison: fixed lag versus lag plus position-free n-grams, and all-byte versus answer-marked writes. All arms use float16 readouts and float64 local updates. Keep the tested fixed-lag all-byte store as the reference. Add a count-matched random-write control for the combined encoder. Use the same streams, target cohort and phase-wise write counts for the two restricted-write schedules. Teach markers appear at feedback time and carry no answer identity.

Use the same physical envelope for every arm. Charge fixed encoder and expansion arrays, readout, cache and scratch buffers. Set the envelope from a measured reference census before sealed runs. Choose the combined encoder width on engineering worlds under that envelope, then freeze it and report any dimension differences. Record peak process RSS and time per byte separately.

Workload and pass criteria

Begin at the 512-fact load with four paired development worlds. Keep the untouched old cohort, genuine revisions and misleading one-off feedback. Probe trained and withheld forms after wave D and after clutter. Prespecify separate shift-one, shift-two, wrapper and moved-key tests. Audit the full training windows for accidental exposures. Score immediate revision at eight presentations, durable correction after later interference, and recovery after misleading feedback separately.

Proposed screen: the combined encoder with teach gating must reach at least 90% post-clutter still-valid recall and 85% on each withheld rendering family, with durable revision at least 90%. Require at least +10 points in mean withheld-rendering recall over its count-matched random-write control. Report per-world scores and paired intervals. Four worlds screen the combination. Confirm a passing configuration on fresh worlds before changing the reference budget.

Integrate once components qualify

First mixed-stream system

Keep three roles separate. A fixed position-tolerant factual pathway learns marked answers. A relational pathway learns structural transitions and uses the selected deployment policy. An all-byte predictor handles the rest of the stream. Route predictions using the visible query prefix and route writes when legitimate feedback arrives. Start with explicit record formats and count the router and all three pathways in the resource census.

Interleave background transitions, factual acquisition and correction, and parity queries with a structural change. Compare each isolated component on the identical mixed-stream exposure schedule against its integrated counterpart. Freeze the routing and permitted writes before evaluation. Report factual recall and durable revision, relational still-valid retention and adaptation, own-answer-plus-newline scores, whole-stream bits per byte, and resource cost separately.

Proposed integration gate: on the same worlds, no more than 5 percentage points loss from isolation on any primary component endpoint and continued attainment of the component floors. Report bits per byte against the all-byte predictor alone. Use routing and write counters to localize cross-path interference before allowing shared mutable features.

Separate follow-up on pathway selection

Once the stability result is interpretable, test a predeclared small bank of temporal strides at a fixed feature and readout budget. Compare a local feedback-driven mixture against a fixed mixture and the best single path chosen on development worlds. Include an asymmetric relation and reverse-role probes. Commit all strides before the evaluation. Treat variable-format operand binding and arithmetic as separate experiments.

Execution record

Seal world seeds, pair rosters, phase budgets, source hashes and gate code before final runs. Use engineering worlds to verify pre-answer scoring, absent feedback for held-out pairs, no state resets, and deterministic reruns. Preserve checkpoint and per-world receipts, scores by panel, and the input and resource accounting needed for independent reduction. This document specifies proposed experiments and thresholds.

Source records

User-provided follow-up audit of MiniFly archive commit 06c60775. The archive contains the clean byte-parity and float16 storage runners and results.